

Name:

Collaborators:

Instructions: Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

Modeling Solution Mixing

1. Mixing Problem

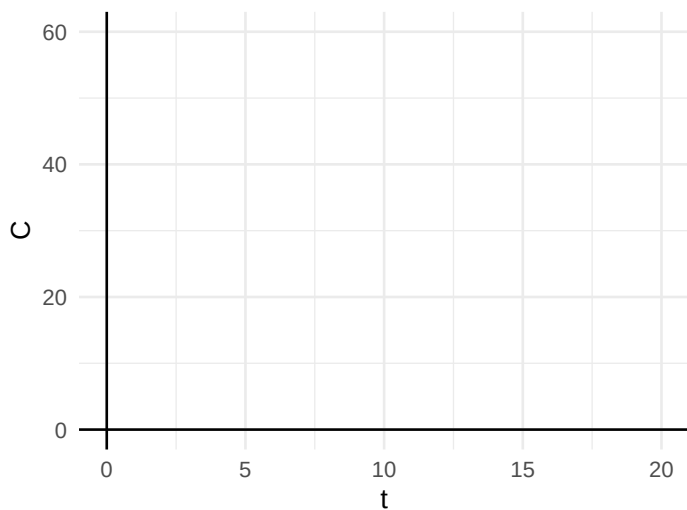
The amount of a substance C in a solution (a mixture of a fluid and a substance) at time t is modeled by the ODE

$$C' = 6 - \frac{2C}{10 + Mt},$$

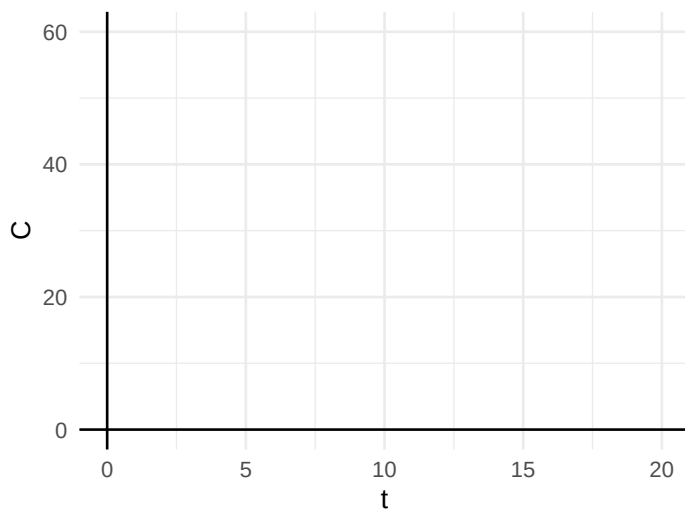
where C' is the substance rate of change, the constant term 6 is the rate of incoming solution, the term $-\frac{2C}{10+t}$ represents the rate at which the substance flows out, with the numerator corresponding to the amount of substance and the denominator representing the fluid volume at time t that changes at a rate of M .

- a. Use the blank axis below to draw complete and well-represented slope fields as the parameter M changes for the cases:

i. $M = -1$



ii. $M = 0$



- b. For each M case in Part (a), describe the general behavior of the substance using the slope fields. Assume that the initial substance starts at $C(0) > 0$.

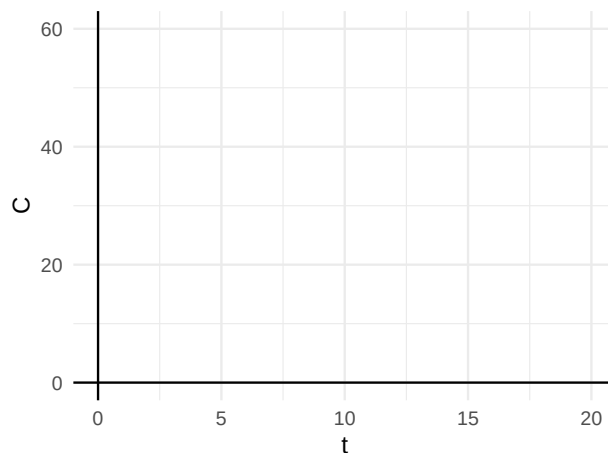
2. Solving a Mixing Problem

Continued from Problem (1). Consider the initial value problem,

$$C' = 6 - \frac{2C}{10 + t}, \quad C(0) = 50.$$

- a. Use the method of integrating factors to determine the general solution.

- c. Use the blank axis below to draw a complete and well represented slope field of the ODE. Then, draw the specific solution on the same field.



- b. Determine the specific solution.

- d. Describe what happens when the inflow stops. That is when the the rate of incoming solution becomes 0.

- e. Describe what happens when the tank overflows. At what point in time does the overflow happen if the tank capacity is 100?